

**TUAS WATER RECLAMATION PLANT
CONTRACT 4B - DIGESTION PRE-TREATMENT AND SIDESTREAM NITROGEN REMOVAL FACILITIES (2ND-STAGE)**

Note: Documents marked with asterisk (*) are issued under 2nd-Stage Tender.

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VOLUME 5 : PART-IA - TENDER DRAWINGS

Project Prefix	Facility Name	Disc Code	Drawing Type	Drawing Number	Drawing Description	Revision	Remarks
Architectural							
WSW41	C4B-Site wide	A	General (symbols, legends, general notes, etc.)	131101-WSW41-DR-A-0001	OVERALL GENERAL, CONSTRUCTION NOTES FIRE LEGEND	T3	*
WSW41	C4B-Site wide	A	Details	131101-WSW41-DR-A-5001	ARCHITECTURAL STANDARD DETAILS PARAPET AND ROOF DRAINAGE	T1	
WSW41	C4B-Site wide	A	Details	131101-WSW41-DR-A-5002	ARCHITECTURAL STANDARD DETAILS SIPHONIC OUTLET AND SPLASH BLOCK	T1	
WSW41	C4B-Site wide	A	Details	131101-WSW41-DR-A-5003	ARCHITECTURAL STANDARD DETAILS DISCHARGE AND WALL PENETRATION	T1	
WSW41	C4B-Site wide	A	Details	131101-WSW41-DR-A-5004	ARCHITECTURAL FIRE ESCAPE CAT LADDER & INTERMEDIATE	T1	
WSW41	C4B-Site wide	A	Details	131101-WSW41-DR-A-5005	ARCHITECTURAL STANDARD DETAILS PARAPET, GROOVES AND CAT LADDER	T1	
WSW41	C4B-Site wide	A	Details	131101-WSW41-DR-A-5006	ARCHITECTURAL STANDARD DETAILS RAILINGS	T1	
WSW41	C4B-Site wide	A	Details	131101-WSW41-DR-A-5007	ARCHITECTURAL STANDARD DETAILS CONVENTIONAL OUTLET	T1	
WSW41	C4B-Site wide	A	Details	131101-WSW41-DR-A-5101	DOOR TYPES	T1	
Building Services							
WSD11	Digestion Pre-Treatment 1	B	Schematics, Schedules and Diagrams	131101-WSD11-DR-B-7701	DIGESTION PRE-TREATMENT 1 (WSD11) SCHEMATIC FOR PLUMBING SYSTEM	T1	
WST02	Side Stream Nitrogen Removal	B	Schematics, Schedules and Diagrams	131101-WST02-DR-B-7101	SIDESTREAM NITROGEN REMOVAL (WST02) SCHEMATIC FOR FIRE ALARM PANEL	T1	
WST02	Side Stream Nitrogen Removal	B	Schematics, Schedules and Diagrams	131101-WST02-DR-B-7103	SIDESTREAM NITROGEN REMOVAL (WST02) SCHEMATIC FOR PUBLIC ADDRESS SYSTEM	T1	
WST02	Side Stream Nitrogen Removal	B	Schematics, Schedules and Diagrams	131101-WST02-DR-B-7104	SIDESTREAM NITROGEN REMOVAL (WST02) SCHEMATIC FOR DRY RISER SYSTEM	T2	
WST02	Side Stream Nitrogen Removal	B	Schematics, Schedules and Diagrams	131101-WST02-DR-B-7201	SIDESTREAM NITROGEN REMOVAL (WST02) SCHEMATIC FOR AIR-CONDITIONING & MECHANICAL VENTILATION SYSTEM (ACMV)	T1	
WST02	Side Stream Nitrogen Removal	B	Schematics, Schedules and Diagrams	131101-WST02-DR-B-7301	SIDESTREAM NITROGEN REMOVAL (WST02) SCHEMATIC FOR BMS	T1	
WST02	Side Stream Nitrogen Removal	B	Schematics, Schedules and Diagrams	131101-WST02-DR-B-7501	SIDESTREAM NITROGEN REMOVAL (WST02) SCHEMATIC FOR SECURITY CCTV SYSTEM	T1	
WST02	Side Stream Nitrogen Removal	B	Schematics, Schedules and Diagrams	131101-WST02-DR-B-7502	SIDESTREAM NITROGEN REMOVAL (WST02) SCHEMATIC FOR PLANT CCTV SYSTEM	T1	
WST02	Side Stream Nitrogen Removal	B	Schematics, Schedules and Diagrams	131101-WST02-DR-B-7503	SIDESTREAM NITROGEN REMOVAL (WST02) SCHEMATIC FOR DOOR ACCESS CABLING	T1	
WST02	Side Stream Nitrogen Removal	B	Schematics, Schedules and Diagrams	131101-WST02-DR-B-7701	SIDESTREAM NITROGEN REMOVAL (WST02) SCHEMATIC FOR PLUMBING AND SANITARY SYSTEM	T1	
Civil							
WSW41	C4B-Site wide	C	Plan Sheets, Plan & Profile	131101-WSW41-DR-C-0001	LOCATION & SITE PLAN	T1	
WSW41	C4B-Site wide	C	Plan Sheets, Plan & Profile	131101-WSW41-DR-C-0002	HOARDING PLAN	T1	
WSW41	C4B-Site wide	C	Plan Sheets, Plan & Profile	131101-WSW41-DR-C-0003	SITE UTILISATION PLAN	T1	

Drawing Register
Tender

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VOLUME 5 : PART-IA - TENDER DRAWINGS

Project Prefix	Facility Name	Disc Code	Drawing Type	Drawing Number	Drawing Description	Revision	Remarks
Electrical							
WIL00	Industrial Common	E	Schematics, Schedules and Diagrams	131101-WIL00-DR-E-7000	INDUSTRIAL LM1 SINGLE LINE DIAGRAM OVERALL SHEET 1 OF 13	T1	
WIL00	Industrial Common	E	Schematics, Schedules and Diagrams	131101-WIL00-DR-E-7015	INDUSTRIAL LM1 SINGLE LINE DIAGRAM HV SWITCHGEAR WIL00-HVSG-0001A	T1	
WIL00	Industrial Common	E	Schematics, Schedules and Diagrams	131101-WIL00-DR-E-7016	INDUSTRIAL LM1 SINGLE LINE DIAGRAM HV SWITCHGEAR WIL00-HVSG-0001B	T1	
WSB01	Biosolids Building	E	Schematics, Schedules and Diagrams	131101-WSB01-DR-E-7000	BIOSOLIDS BUILDING SINGLE LINE DIAGRAM OVERALL	T2	*
WSE00	Electrical Common	E	General (symbols legends, general notes, etc)	131101-WSE00-DR-E-0021	ELECTRICAL LEGEND AND ABBREVIATIONS	T1	
WST02	Side Stream Nitrogen Removal	E	Plan Sheets, Plan & Profile	131101-WST02-DR-E-3000	ELECTRICAL DUCT BANKS CROSS-SECTIONS	T1	
WST02	Side Stream Nitrogen Removal	E	Schematics, Schedules and Diagrams	131101-WST02-DR-E-7000	SIDESTREAM N REMOVAL SINGLE LINE DIAGRAM OVERALL	T1	
WST02	Side Stream Nitrogen Removal	E	Schematics, Schedules and Diagrams	131101-WST02-DR-E-7300	SIDESTREAM N REMOVAL SCHEMATIC EARTHING	T1	
WSW41	C4B-Site wide	E	Plan Sheets, Plan & Profile	131101-WSW41-DR-E-1003	DUCTBANKS AND MANHOLES	T1	
WSW41	C4B-Site wide	E	Schematics, Schedules and Diagrams	131101-WSW41-DR-E-7400	CABLE SELECTION CRITERIA	T1	
Instrumentation Control and Automation							
WSD11	Digestion Pre-Treatment 1	I	Schematics, Schedules and Diagrams	131101-WSD11-DR-I-4000	BIOSOLIDS BUILDING THP GENERAL RING PLAN	T1	
WSD11	Digestion Pre-Treatment 1	I	Schematics, Schedules and Diagrams	131101-WSD11-DR-I-7001	CONTROL NETWORK DIAGRAM BIOSOLIDS BUILDING PROCESS XLV ROOM PART-1	T1	
WSD11	Digestion Pre-Treatment 1	I	Schematics, Schedules and Diagrams	131101-WSD11-DR-I-7002	CONTROL NETWORK DIAGRAM BIOSOLIDS BUILDING PROCESS XLV ROOM PART-2	T2	*
WSD11	Digestion Pre-Treatment 1	I	Schematics, Schedules and Diagrams	131101-WSD11-DR-I-7003	CONTROL NETWORK DIAGRAM THP FIELD & MCC	T1	
WSD11	Digestion Pre-Treatment 1	I	Schematics, Schedules and Diagrams	131101-WSD11-DR-I-7004	CONTROL NETWORK DIAGRAM THP TRAINS LCP	T2	*
WSD11	Digestion Pre-Treatment 1	I	Schematics, Schedules and Diagrams	131101-WSD11-DR-I-7030	POWER SUPPLY TO MCS AT BIOSOLIDS BUILDING PART 1	T2	*
WSD11	Digestion Pre-Treatment 1	I	Schematics, Schedules and Diagrams	131101-WSD11-DR-I-7031	POWER SUPPLY TO MCS AT BIOSOLIDS BUILDING PART 2	T1	
WST02	Side Stream Nitrogen Removal	I	Schematics, Schedules and Diagrams	131101-WST02-DR-I-4000	SIDESTREAM N REMOVAL RING PLAN	T1	
WST02	Side Stream Nitrogen Removal	I	Schematics, Schedules and Diagrams	131101-WST02-DR-I-7001	CONTROL NETWORK DIAGRAM SIDESTREAM N REMOVAL PROCESS XLV ROOM	T1	
WST02	Side Stream Nitrogen Removal	I	Schematics, Schedules and Diagrams	131101-WST02-DR-I-7010	CONTROL NETWORK DIAGRAM SIDESTREAM N REMOVAL FIELD AND MCC	T1	
WST02	Side Stream Nitrogen Removal	I	Schematics, Schedules and Diagrams	131101-WST02-DR-I-7011	CONTROL NETWORK DIAGRAM AERATION BLOWER FIELD	T1	
WST02	Side Stream Nitrogen Removal	I	Schematics, Schedules and Diagrams	131101-WST02-DR-I-7030	POWER SUPPLY TO MCS AT SIDESTREAM N REMOVAL	T1	
WSW00	Site Overall	I	General (symbols legends, general notes, etc)	131101-WSW00-DR-I-0010	GENERAL NOTES	T1	

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VOLUME 5 : PART-IA - TENDER DRAWINGS

Project Prefix	Facility Name	Disc Code	Drawing Type	Drawing Number	Drawing Description	Revision	Remarks
WSW00	Site Overall	I	General (symbols legends, general notes, etc)	131101-WSW00-DR-I-0011	GENERAL LEGEND AND ABBREVIATIONS PROCESS	T1	
WSW00	Site Overall	I	General (symbols legends, general notes, etc)	131101-WSW00-DR-I-0012	GENERAL LEGEND AND ABBREVIATIONS NON-PROCESS	T1	
WSW00	Site Overall	I	Schematics, Schedules and Diagrams	131101-WSW00-DR-I-5811	ICA F/O CABLE DUCTS	T1	
WSW00	Site Overall	I	Schematics, Schedules and Diagrams	131101-WSW00-DR-I-7727	TYPICAL PMS INTERFACE SCHEMATIC - LOW VOLTAGE SWITCHBOARDS, MCCS AND GENSET	T1	
Process							
WIL11	Industrial LM1 Headworks	P	Schematics, Schedules and Diagrams	131101-WIL11-DR-P-6000	P&ID INTERMEDIATE SCREENS 1, 2 & 3	T1	
WIL11	Industrial LM1 Headworks	P	Schematics, Schedules and Diagrams	131101-WIL11-DR-P-6008	P&ID GRIT TANKS 5 & 6	T1	
WIL12	Industrial LM1 Primary Treatment	P	Schematics, Schedules and Diagrams	131101-WIL12-DR-P-6023	P&ID PSI TANKS AND TRANSFER PUMPS	T1	
WIL13	Industrial LM1 Secondary Treatment	P	Schematics, Schedules and Diagrams	131101-WIL13-DR-P-6042	P&ID WAS PUMPING NO. 2	T1	
WIL13	Industrial LM1 Secondary Treatment	P	Schematics, Schedules and Diagrams	131101-WIL13-DR-P-6080	P&ID MBR FILTRATE TANK & TRANSFER PUMPS PART 2	T1	
WIL22	Industrial LM2 Primary Treatment	P	Schematics, Schedules and Diagrams	131101-WIL22-DR-P-6023	P&ID PSI TANKS AND TRANSFER PUMPS	T1	
WIL23	Industrial LM2 Secondary Treatment	P	Schematics, Schedules and Diagrams	131101-WIL23-DR-P-6042	P&ID WAS PUMPING NO. 2	T1	
WIL23	Industrial LM2 Secondary Treatment	P	Schematics, Schedules and Diagrams	131101-WIL23-DR-P-6080	P&ID MBR FILTRATE TANK & TRANSFER PUMPS PART 2	T1	
WSB01	Biosolids Building	P	Schematics, Schedules and Diagrams	131101-WSB01-DR-P-6010	P&ID PSI HOLDING TANK AND CIRCULATION PUMPS	T1	
WSB01	Biosolids Building	P	Schematics, Schedules and Diagrams	131101-WSB01-DR-P-6020	P&ID WAS HOLDING TANK AND CIRCULATION PUMPS	T1	
WSB01	Biosolids Building	P	Schematics, Schedules and Diagrams	131101-WSB01-DR-P-6026	P&ID DWAS SILO 1 & FEED CONVEYORS	T1	
WSB01	Biosolids Building	P	Schematics, Schedules and Diagrams	131101-WSB01-DR-P-6027	P&ID DWAS SILO 2 & FEED CONVEYORS	T1	
WSB01	Biosolids Building	P	Schematics, Schedules and Diagrams	131101-WSB01-DR-P-6080	P&ID TCD WETWELL AND PUMPS	T1	
WSB01	Biosolids Building	P	Schematics, Schedules and Diagrams	131101-WSB01-DR-P-6081	P&ID TCI WETWELL AND PUMPS	T1	
WSB01	Biosolids Building	P	Schematics, Schedules and Diagrams	131101-WSB01-DR-P-6082	P&ID DC WETWELL AND PUMPS	T1	
WSB01	Biosolids Building	P	Schematics, Schedules and Diagrams	131101-WSB01-DR-P-6090	P&ID PLANT DRAIN PUMPING STATION	T1	
WSB01	Biosolids Building	P	Schematics, Schedules and Diagrams	131101-WSB01-DR-P-6200	P&ID BIOSOLIDS BUILDING G2D NETWORK	T1	
WSB01	Biosolids Building	P	Schematics, Schedules and Diagrams	131101-WSB01-DR-P-6300	P&ID BIOSOLIDS BUILDING PD NETWORK	T1	
WSB01	Biosolids Building	P	Schematics, Schedules and Diagrams	131101-WSB01-DR-P-6600	P&ID BIOSOLIDS BUILDING IA NETWORK	T1	
WSB01	Biosolids Building	P	Schematics, Schedules and Diagrams	131101-WSB01-DR-P-7001	PFD BIOSOLIDS PHASE-1	T1	

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WSD00	Digestion Common	P	Schematics, Schedules and Diagrams	131101-WSD00-DR-P-7001	PFD BIOSOLIDS MASS/HEAT BALANCE ANNUAL AVERAGE	T1	
WSD00	Digestion Common	P	Schematics, Schedules and Diagrams	131101-WSD00-DR-P-7002	PFD BIOSOLIDS MASS/HEAT BALANCE MONTHLY MAXIMUM	T1	
WSD02	Digestion	P	Schematics, Schedules and Diagrams	131101-WSD02-DR-P-6100	P&ID DIGESTERS OVERALL	T1	
WSD05	Greasy Waste Receiving Facility	P	Schematics, Schedules and Diagrams	131101-WSD05-DR-P-6021	P&ID GW TRANSFER PIPE TO GW HOLDING TANK	T1	
WSD11	Digestion Pre-Treatment 1	P	Schematics, Schedules and Diagrams	131101-WSD11-DR-P-6000	P&ID THP FEED PUMPS	T2	*
WSD11	Digestion Pre-Treatment 1	P	Schematics, Schedules and Diagrams	131101-WSD11-DR-P-6010	P&ID THERMAL HYDROLYSIS PROCESS PACKAGE 1	T2	*
WSD11	Digestion Pre-Treatment 1	P	Schematics, Schedules and Diagrams	131101-WSD11-DR-P-6020	P&ID THERMAL HYDROLYSIS PROCESS PACKAGE 2	T2	*
WSD11	Digestion Pre-Treatment 1	P	Schematics, Schedules and Diagrams	131101-WSD11-DR-P-6030	P&ID THERMAL HYDROLYSIS PROCESS PACKAGE 3	T2	*
WSD11	Digestion Pre-Treatment 1	P	Schematics, Schedules and Diagrams	131101-WSD11-DR-P-6040	P&ID THERMAL HYDROLYSIS PROCESS PACKAGE 4	T2	*
WSD11	Digestion Pre-Treatment 1	P	Schematics, Schedules and Diagrams	131101-WSD11-DR-P-6050	P&ID THP STHP DISTRIBUTION	T2	*
WSD11	Digestion Pre-Treatment 1	P	Schematics, Schedules and Diagrams	131101-WSD11-DR-P-6060	P&ID THP 1 HEAT EXCHANGERS	T1	
WSD11	Digestion Pre-Treatment 1	P	Schematics, Schedules and Diagrams	131101-WSD11-DR-P-6061	P&ID THP 2 HEAT EXCHANGERS	T1	
WSD11	Digestion Pre-Treatment 1	P	Schematics, Schedules and Diagrams	131101-WSD11-DR-P-6062	P&ID THP 3 HEAT EXCHANGERS	T2	*
WSD11	Digestion Pre-Treatment 1	P	Schematics, Schedules and Diagrams	131101-WSD11-DR-P-6063	P&ID THP 4 HEAT EXCHANGERS	T2	*
WSD11	Digestion Pre-Treatment 1	P	Schematics, Schedules and Diagrams	131101-WSD11-DR-P-6070	P&ID HS DIGESTER FEED PUMPS	T2	*
WSD11	Digestion Pre-Treatment 1	P	Schematics, Schedules and Diagrams	131101-WSD11-DR-P-6200	P&ID THP AREA G2D NETWORK	T2	*
WSD11	Digestion Pre-Treatment 1	P	Schematics, Schedules and Diagrams	131101-WSD11-DR-P-6600	P&ID THP AREA INSTRUMENT AIR NETWORK	T1	
WST02	Side Stream Nitrogen Removal	P	Schematics, Schedules and Diagrams	131101-WST02-DR-P-6000	P&ID EQUALIZATION TANKS	T1	
WST02	Side Stream Nitrogen Removal	P	Schematics, Schedules and Diagrams	131101-WST02-DR-P-6001	P&ID SIDESTREAM - N REMOVAL DEAMMONIFICATION REACTORS 1&2	T1	
WST02	Side Stream Nitrogen Removal	P	Schematics, Schedules and Diagrams	131101-WST02-DR-P-6002	P&ID SIDESTREAM - N REMOVAL DEAMMONIFICATION REACTORS 3&4	T1	
WST02	Side Stream Nitrogen Removal	P	Schematics, Schedules and Diagrams	131101-WST02-DR-P-6003	P&ID SIDESTREAM - N REMOVAL DISCHARGE AND SLUDGE WET WELL AND TRANSFER PUMPS	T2	*
WST02	Side Stream Nitrogen Removal	P	Schematics, Schedules and Diagrams	131101-WST02-DR-P-6004	P&ID SSN AERATION BLOWER (1 TO 4)	T1	
WST02	Side Stream Nitrogen Removal	P	Schematics, Schedules and Diagrams	131101-WST02-DR-P-6005	P&ID SSN AERATION BLOWER (5 & 6)	T2	*
WST02	Side Stream Nitrogen Removal	P	Schematics, Schedules and Diagrams	131101-WST02-DR-P-7001	PFD SIDESTREAM N REMOVAL	T1	
WSW00	Site Overall	P	General (symbols, legends, general notes, etc.)	131101-WSW00-DR-P-0001	P&ID LEGEND GENERAL	T1	

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WSW00	Site Overall	P	General (symbols, legends, general notes, etc.)	131101-WSW00-DR-P-0002	P&ID LEGEND TAGGING CONVENTION	T1	
WSW00	Site Overall	P	General (symbols, legends, general notes, etc.)	131101-WSW00-DR-P-0003	P&ID LEGEND TAGGING METADATA	T1	
WSW00	Site Overall	P	General (symbols, legends, general notes, etc.)	131101-WSW00-DR-P-0004	P&ID LEGEND INSTRUMENT IDENTIFICATION	T1	
WSW00	Site Overall	P	General (symbols, legends, general notes, etc.)	131101-WSW00-DR-P-0005	P&ID LEGEND SYMBOLS SHEET 1 OF 2	T1	
WSW00	Site Overall	P	General (symbols, legends, general notes, etc.)	131101-WSW00-DR-P-0006	P&ID LEGEND SYMBOLS SHEET 2 OF 2	T1	
WSW00	Site Overall	P	General (symbols, legends, general notes, etc.)	131101-WSW00-DR-P-0014	P&ID LEGEND LOCAL CONTROL STATION SHEET 1 OF 4	T1	
WSW00	Site Overall	P	General (symbols, legends, general notes, etc.)	131101-WSW00-DR-P-0015	P&ID LEGEND LOCAL CONTROL STATION SHEET 2 OF 4	T1	
WSW00	Site Overall	P	General (symbols, legends, general notes, etc.)	131101-WSW00-DR-P-0016	P&ID LEGEND LOCAL CONTROL STATION SHEET 3 OF 4	T1	
WSW00	Site Overall	P	General (symbols, legends, general notes, etc.)	131101-WSW00-DR-P-0017	P&ID LEGEND LOCAL CONTROL STATION SHEET 4 OF 4	T1	
Structural							
WSW41	C4B-Site wide	S	General (symbols, legends, general notes, etc.)	131101-WSW41-DR-S-1001	NOTIONAL PIPE RACK PLAN	T1	
WSW41	C4B-Site wide	S	General (symbols, legends, general notes, etc.)	131101-WSW41-DR-S-3001	NOTIONAL PIPE RACK SECTIONS, ELEVATION AND STUMP DETAILS	T1	
WSW41	C4B-Site wide	S	General (symbols, legends, general notes, etc.)	131101-WSW41-DR-S-5001	STANDARD DETAILS INTERNAL PROTECTION COATING	T1	
Total No. of Drawings						106	